

Complete mappings of the Brauer and partial Brauer monoids

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Abstract

A complete mapping of a semigroup S is a bijection $\alpha: S \rightarrow S$ such that $x \mapsto x \cdot x\alpha$ is also a bijection. Problem 15.11 of Araújo, Bentz, Cameron, Hendrey and Kinyon (arXiv:2608.25092) asks for a classification of complete mappings of the Brauer monoid $\mathcal{B}_n$ and the partial Brauer monoid $\mathcal{PB}_n$. By their reduction to principal factors and the Hall–Paige theorem, the open cases are the principal factors of ranks 2 and 3, together with the question of which of the monoids have a complete mapping. We settle the existence question in all cases. Every proper principal factor of $\mathcal{B}_n$ has a complete mapping. Every proper principal factor of $\mathcal{PB}_n$ has one, except the rank-2 factor of $\mathcal{PB}_3$, which is the Brandt semigroup $B(S_2, 3)$ and has none. Consequently $\mathcal{B}_n$ and $\mathcal{PB}_n$ have a complete mapping if and only if $n \notin \{2, 3\}$, as for the full transformation monoid and the partition monoid. The new case is that of $\mathcal{B}_n$ with $n \equiv 2, 3 \pmod{4}$, in which the rank-2 or rank-3 factor has an odd number of $\mathcal{L}$ -classes. For it we give an explicit complete mapping, built from a triangle of half-diagrams whose sandwich permutations have odd product and from a perfect matching of the remaining half-diagrams. The other cases follow directly from results of Araújo et al. once the Rees structure of the principal factors is written down. The two explicit constructions of complete mappings of Rees 0-matrix semigroups over $\mathbb{Z}_2$ on which the proofs rest are formally verified in Lean 4, for arbitrary finite index sets; the rest of the argument is not formalised. We also give explicit complete mappings of $\mathcal{B}_n$ and $\mathcal{PB}_n$ for $n \in \{1, 4, 5, 6\}$ and of the two smallest factors in the new case, each checked by a program that shares no code with the program that built it. The proofs were found by AI coding agents working under the author’s direction; their role is described at the end of the paper.

1 Introduction

Background. Let $(S, \cdot)$ be a magma. A *complete mapping* of S is a bijection $\alpha: S \rightarrow S$ such that the map $\theta: S \rightarrow S$, $x\theta = x \cdot x\alpha$, is also a bijection. As in [1], maps act on the right. For a finite magma this is the same as a transversal of the Cayley table. For finite groups, existence is decided by the Hall–Paige theorem, in the form recorded as [1, Theorem 1.1]: a finite group has a complete mapping if and only if its Sylow 2-subgroups are trivial or non-cyclic. In particular, the symmetric group S_r has a complete mapping if and only if $r \notin \{2, 3\}$.

Araújo, Bentz, Cameron, Hendrey and Kinyon [1] developed the theory for finite semigroups. A finite semigroup has a complete mapping if and only if each of its principal factors has one [1, Theorem 4.4], which reduces the question to Rees 0-matrix semigroups; these are studied in detail in [1, Sections 6 and 7]. Among the applications, the full transformation monoid T_n and the partition monoid $\mathcal{P}_n$ have complete mappings if and only if $n = 1$ or $n \geq 4$ [1, Theorems 9.16 and 8.5], the symmetric inverse monoid I_n has one if and only if $n \equiv 0, 1$

(mod 4) [1, Corollary 10.2], and the planar partition, Motzkin and Jones monoids have complete mappings [1, Corollary 8.8]. The Brauer monoid $\mathcal{B}_n$ and the partial Brauer monoid $\mathcal{PB}_n$ were left open. Problem 15.11 of [1] asks for a classification of their complete mappings; it notes that, by the Hall–Paige theorem, the proper principal factors still to be decided are those of ranks 2 and 3, and it also asks for which n the whole monoid has a complete mapping. Here a principal factor is *proper* if it is not the principal factor of the group of units.

Results. We decide every case.

Theorem 1.1. *Let $n \geq 1$.*

- (a) *Every proper principal factor of $\mathcal{B}_n$ has a complete mapping.*
- (b) *Every proper principal factor of $\mathcal{PB}_n$ has a complete mapping, except the principal factor of rank 2 of $\mathcal{PB}_3$. That factor is isomorphic to the Brandt semigroup $B(S_2, 3) = \mathcal{M}^0(S_2, [3], [3], \Delta_3)$, where Δ_3 is the 3×3 identity matrix, and it has no complete mapping.*
- (c) *The following are equivalent: (i) $\mathcal{B}_n$ has a complete mapping; (ii) $\mathcal{PB}_n$ has a complete mapping; (iii) S_n has a complete mapping; (iv) $n = 1$ or $n \geq 4$.*

The principal factor of rank r is a Rees 0-matrix semigroup $\mathcal{M}^0(S_r, H_r, H_r, P)$ over the symmetric group S_r , indexed on both sides by the set H_r of half-diagrams of rank r (Proposition 3.1). Its sandwich matrix has identity diagonal. So when S_r has a complete mapping, that is, when $r \notin \{2, 3\}$, the factor has one by [1, Theorem 6.2]. When $r \in \{2, 3\}$ and $N = |H_r|$ is even, it has one by [1, Corollary 7.6]. The remaining factors are those with $r \in \{2, 3\}$ and N odd, the case in which both index sets are odd and, as [1, Section 7] explains, the images of the sandwich entries in a cyclic 2-quotient of S_r matter, not just the pattern of non-zero entries. Table 1 lists the parities. For $\mathcal{PB}_n$ the only proper factor with $r \in \{2, 3\}$ and N odd is the rank-2 factor of $\mathcal{PB}_3$, and it has no complete mapping by [1, Theorem 7.19]. For $\mathcal{B}_n$ there are infinitely many such factors, those of rank 2 for $n \equiv 2 \pmod{4}$ and of rank 3 for $n \equiv 3 \pmod{4}$; the first are the rank-2 factor of $\mathcal{B}_6$, with $N = 45$ and 4050 non-zero elements, and the rank-3 factor of $\mathcal{B}_7$, with $N = 105$ and 66150 non-zero elements. For these factors we construct complete mappings explicitly (Section 5), from three pairwise compatible half-diagrams whose sandwich permutations have odd product and a perfect matching of the other half-diagrams by compatible pairs.

Monoid	rank r	$N = H_r $	N odd exactly when	complete mapping of the factor
$\mathcal{B}_n$	2	$\binom{n}{2} (n-3)!!$	$n \equiv 2 \pmod{4}$	yes: [1, Corollary 7.6] if N is even, Section 5 if N is odd
$\mathcal{B}_n$	3	$\binom{n}{3} (n-4)!!$	$n \equiv 3 \pmod{4}$	yes: as for rank 2
$\mathcal{PB}_n$	2, 3	$\binom{n}{r} a(n-r)$	$(n, r) = (3, 2)$	yes if N is even [1, Corollary 7.6]; none for $(n, r) = (3, 2)$ (Proposition 6.2)

Table 1: The proper principal factors of ranks 2 and 3 ($r < n$, and $r \equiv n \pmod{2}$ for $\mathcal{B}_n$). Here N is the number of $\mathcal{L}$ -classes of the factor, which equals the number of $\mathcal{R}$ -classes, and $a(k)$ is the number of involutions of a k -set. The factors of rank $r \notin \{2, 3\}$ have complete mappings by [1, Theorem 6.2].

All our complete mappings are given by explicit formulas (Lemmas 4.1–4.4). The two constructions for ranks 2 and 3 (Lemmas 4.3 and 4.4), stated for a Rees 0-matrix semigroup over $\mathbb{Z}_2$ with identity diagonal and an arbitrary finite index set, are formally verified in Lean 4

(Section 9). The facts about diagrams (Sections 3, 5 and 6) and the results of [1] that we use are not formalised; for them the evidence is the written proofs, supported by the computer checks of Section 8.

Scope. We read “classify complete mappings” in the sense explained by the second sentence of Problem 15.11: decide, for every n and every rank r , whether the principal factor of rank r has a complete mapping, and decide whether the whole monoid has one. Theorem 1.1 does this completely. We do not describe the set of all complete mappings, as [1, Proposition 6.5] does for the combinatorial Brandt semigroups by means of Latin squares.

Prior work. Problem 15.11 appears in version 1 of [1], submitted to the arXiv on 25 August 2026, which was still the only version when we checked on 26 September 2026. We found no solution of, and no announced progress on, Problem 15.11 in the following sources, which we searched on 26 September 2026 between 06:49 and 07:46 UTC: the arXiv (the listing of [1] and eight title and abstract searches through the arXiv API); GitHub searches of issues, pull requests, commits, code and repositories; the public repositories of AI-assisted work on open problems that we check, namely the file tree, issues and pull requests of the `trureturing` repository [7], and the repositories SCOPE2026, AI-Has-Taste, formal-conjectures and astrafala; the problem database MathDB (<https://mathdb.com>); MathOverflow and Mathematics Stack Exchange, through their search interface; Crossref; web search; the blog post announcing [1] and its comments [2], and the publication list and arXiv listing of the third author; and the slides of a 2025 talk on the subject by the fifth author [5]. The only earlier AI-assisted work on [1] that we found is issue #9377 and pull request #9405 of [7], which concern Problem 15.5 and do not mention Problem 15.11 or diagram monoids. Our searches on 26 September 2026 for works citing [1] in OpenAlex and Semantic Scholar failed (HTTP error 429, rate limit exceeded), so citing works were not checked systematically; we did not search Google Scholar, zbMATH or MathSciNet. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, where [1] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. This reports the coverage of our searches, not a guarantee of priority. Our results are a direct extension of the methods of [1]: the reduction to principal factors, the criteria for Rees 0-matrix semigroups, the lifting lemma and the non-existence theorem that we use are theirs, and our treatment of $\mathcal{B}_n$ follows their proof for the partition monoid [1, Theorem 8.5], including the triangle of half-diagrams in Lemma 5.2.

Organisation. Section 2 fixes notation and lists the results of [1] that we use. Section 3 gives the Rees structure of the principal factors of $\mathcal{B}_n$ and $\mathcal{PB}_n$. Section 4 gives explicit complete mappings of Rees 0-matrix semigroups with identity diagonal. Section 5 treats $\mathcal{B}_n$ and proves Theorem 1.1(a), Section 6 treats $\mathcal{PB}_n$ and proves Theorem 1.1(b), and Section 7 proves Theorem 1.1(c). Section 8 reports the computations and Section 9 the formal verification.

2 Preliminaries

Rees 0-matrix semigroups. We follow [1, Section 4]. Let G be a finite group, let I and Λ be non-empty finite sets, and let $P = (p_{\lambda i})$ be a $\Lambda \times I$ matrix over $G \cup \{0\}$ with no zero row or column. The Rees 0-matrix semigroup $\mathcal{M}^0(G, I, \Lambda, P)$ is $(I \times G \times \Lambda) \cup \{0\}$ with the product

$$(i, a, \lambda)(j, b, \mu) = \begin{cases} (i, a p_{\lambda j} b, \mu) & \text{if } p_{\lambda j} \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and $0x = x0 = 0$. The *pattern* of P is the zero-one matrix obtained by replacing every element of G by 1. For a $\mathcal{J}$ -class J of a finite semigroup S , the principal factor J^0 is $J \cup \{0\}$, with the

product of S when it lies in J and 0 otherwise.

Results of [1] that we use. We use the following results, with the numbering of [1].

- (R1) [1, Theorem 1.1] (Hall–Paige). A finite group has a complete mapping if and only if its Sylow 2-subgroups are trivial or non-cyclic.
- (R2) [1, Proposition 1.4]. A complete mapping of a magma with zero fixes the zero, and so does the corresponding map θ .
- (R3) [1, Corollary 3.4]. If a finite monoid has a complete mapping, then so does its group of units.
- (R4) [1, Theorem 4.4]. A finite semigroup has a complete mapping if and only if every principal factor J^0 has one.
- (R5) [1, Lemma 6.3], which extends the lifting construction of [1, Theorem 5.2] to Rees 0-matrix semigroups. Let N be a normal subgroup of G that has a complete mapping, and let $\bar{P}$ be the entrywise image of P in G/N , zero entries staying zero. If $\mathcal{M}^0(G/N, I, \Lambda, \bar{P})$ has a complete mapping, then so does $\mathcal{M}^0(G, I, \Lambda, P)$.
- (R6) [1, Theorem 6.2], implication (c) $\Rightarrow$ (a). If G has a complete mapping and, for every s rows of the pattern, at least $s|I|/|\Lambda|$ columns have a non-zero entry in one of these rows, then $\mathcal{M}^0(G, I, \Lambda, P)$ has a complete mapping. If $|I| = |\Lambda|$ and the pattern contains a one-transversal (a permutation matrix), the condition holds.
- (R7) [1, Corollary 7.6]. If G has no complete mapping, $|I| = |\Lambda|$ is even and the pattern of P contains a one-transversal, then $\mathcal{M}^0(G, I, \Lambda, P)$ has a complete mapping.
- (R8) [1, Theorem 7.19]. If G has non-trivial cyclic Sylow 2-subgroups, $|I|$ and $|\Lambda|$ are odd, and P is obtained from a normalised matrix over G in which every entry has odd order by replacing some entries by 0, then $\mathcal{M}^0(G, I, \Lambda, P)$ has no complete mapping.

By (R1), S_r has a complete mapping for $r \leq 1$ (the trivial group) and for $r \geq 4$ (its Sylow 2-subgroups contain $\langle (1\ 2), (3\ 4) \rangle \cong C_2 \times C_2$), and none for $r \in \{2, 3\}$ (the Sylow 2-subgroups have order 2).

Diagrams. Let $[n] = \{1, \dots, n\}$ and $[n]' = \{1', \dots, n'\}$. The partial Brauer monoid $\mathcal{PB}_n$ consists of the partitions of $[n] \cup [n]'$ into blocks of size 1 or 2, and the Brauer monoid $\mathcal{B}_n$ of those with all blocks of size 2; see [4, Section 2.2]. We draw $[n]$ as the top row and $[n]'$ as the bottom row. The product xy is formed by drawing x above y , identifying the bottom row of x with the top row of y (the *middle row*), keeping the connectivity of the outer points, and discarding closed loops and components contained in the middle row. A block meeting both rows is a *through block*; the *rank* of x is its number of through blocks, and $\text{rank}(xy) \leq \min(\text{rank } x, \text{rank } y)$. As recalled in [1, Section 15], it is proved in [4, Proposition 2.1 and Remark 2.2(v)–(vii)] that the $\mathcal{J}$ -classes of $\mathcal{B}_n$ and $\mathcal{PB}_n$ are the rank classes and that a maximal subgroup in rank r is isomorphic to S_r . We write J_r for the set of elements of rank r ; in $\mathcal{B}_n$ it is non-empty exactly when $r \leq n$ and $r \equiv n \pmod{2}$. The group of units is J_n , the set of permutation diagrams, isomorphic to S_n .

3 Rees coordinates

Let $D \in \{\mathcal{B}_n, \mathcal{PB}_n\}$ and let r be a rank that occurs in D . A *half-diagram* of rank r is a pair $h = (F, M)$, where $F = F(h) \subseteq [n]$ has $|F| = r$ (the *free points*) and $M = M(h)$ is a set of

disjoint 2-subsets (*arcs*) of $[n] \setminus F$ covering $[n] \setminus F$ if $D = \mathcal{B}_n$, and arbitrary if $D = \mathcal{PB}_n$; the points of $[n] \setminus F$ not covered by M are *singletons*. Let $H = H_r(D)$ be the set of half-diagrams of rank r and $N = |H|$. Then

$$N = \binom{n}{r} (n - r - 1)!! \text{ for } \mathcal{B}_n, \quad N = \binom{n}{r} a(n - r) \text{ for } \mathcal{PB}_n,$$

where $(-1)!! = 1$ and $a(k)$ is the number of involutions of a k -set.

For $x \in J_r$, let $\text{top}(x) = (F, M)$, where F is the set of top points of through blocks and M the set of blocks contained in the top row; let $\text{bot}(x)$ be defined in the same way from the bottom row, with primes removed. If $a_1 < \dots < a_r$ are the top points and $b_1 < \dots < b_r$ the bottom points (without primes) of the through blocks, define $g(x) \in S_r$ by requiring that $\{a_k, b'_{kg(x)}\}$ is a block of x for every k . Permutations act on the right, and gh means first g , then h .

For $\lambda, i \in H$, let $\Gamma(\lambda, i)$ be the multigraph on $[n]$ whose edges are the arcs of λ and the arcs of i (an arc of both gives a double edge). Every vertex has degree at most 2, so the components are paths and cycles. A point of $F(\lambda)$ lies on no arc of λ , so it has degree at most 1 and is an end of a path. We say that λ and i are *compatible*, and write $\lambda \sim i$, if the path starting at each point of $F(\lambda)$ ends at a point of $F(i)$; a point of $F(\lambda) \cap F(i)$ is a path of length 0. The r paths are distinct components, so they then match $F(\lambda)$ bijectively with $F(i)$, and we define $p_{\lambda i} \in S_r$ by requiring that the k -th smallest point of $F(\lambda)$ is joined to the $(kp_{\lambda i})$ -th smallest point of $F(i)$.

Proposition 3.1. *Let $D \in \{\mathcal{B}_n, \mathcal{PB}_n\}$, let r be a rank that occurs in D , and let $H = H_r(D)$.*

- (a) *The map $x \mapsto (\text{top}(x), g(x), \text{bot}(x))$ is a bijection from J_r to $H \times S_r \times H$.*
- (b) *For $x, y \in J_r$ we have $\text{rank}(xy) = r$ if and only if $\text{bot}(x) \sim \text{top}(y)$, and then $\text{top}(xy) = \text{top}(x)$, $\text{bot}(xy) = \text{bot}(y)$ and $g(xy) = g(x) p_{\text{bot}(x), \text{top}(y)} g(y)$.*
- (c) *For all $\lambda, i \in H$: $\lambda \sim \lambda$ and $p_{\lambda\lambda} = 1$; and $\lambda \sim i$ if and only if $i \sim \lambda$, in which case $p_{i\lambda} = p_{\lambda i}^{-1}$.*
- (d) *If $F(\lambda) = F(i)$, then $\lambda \sim i$ and $p_{\lambda i} = 1$.*

Consequently $J_r^0 \cong \mathcal{M}^0(S_r, H, H, P)$, where P is the $H \times H$ matrix with entries $p_{\lambda i}$ if $\lambda \sim i$ and 0 otherwise; the isomorphism extends the map in (a) by $0 \mapsto 0$.

Proof. (a) An element of rank r is determined by its blocks inside the top row, its blocks inside the bottom row, and a bijection from its top through points to its bottom through points, and every such choice occurs. The first two data are $\text{top}(x)$ and $\text{bot}(x)$, and $g(x)$ encodes the bijection.

(b) The blocks of x inside the top row are blocks of xy . Consider the top point a_k of a through block of x . Its strand reaches the middle row at $b = b_{kg(x)} \in F(\text{bot}(x))$ and continues along the path of $\Gamma(\text{bot}(x), \text{top}(y))$ starting at b . This path alternates between arcs of $\text{top}(y)$ and arcs of $\text{bot}(x)$, beginning with an arc of $\text{top}(y)$, and stops at a point e at which the next arc is missing. So either $e \in F(\text{top}(y))$, and the strand continues through y to the bottom row; or $e \neq b$ and $e \in F(\text{bot}(x))$, and a_k is joined to another top through point of x ; or e is a singleton of $\text{bot}(x)$ or of $\text{top}(y)$ (only in $\mathcal{PB}_n$), and a_k becomes a singleton. Every through block of xy contains a top point of a through block of x , so $\text{rank}(xy) = r$ if and only if the first case occurs for all k , that is, if and only if $\text{bot}(x) \sim \text{top}(y)$. In that case the blocks of xy inside the top row are exactly those of x , so $\text{top}(xy) = \text{top}(x)$. The r strands end at r distinct bottom through points of y , that is, at all of them, so the blocks of xy inside the bottom row are those of y and $\text{bot}(xy) = \text{bot}(y)$. Following a_k : it reaches the $(kg(x))$ -th smallest point of $F(\text{bot}(x))$, which

is joined to the $(k g(x) p)$ -th smallest point of $F(\text{top}(y))$, where $p = p_{\text{bot}(x), \text{top}(y)}$, and this point is joined in y to the $(k g(x) p g(y))$ -th smallest bottom through point of y .

(c) In $\Gamma(\lambda, \lambda)$ each arc of λ is a double edge and each point of $F(\lambda)$ is isolated, which gives $\lambda \sim \lambda$ and $p_{\lambda\lambda} = 1$. The graphs $\Gamma(\lambda, i)$ and $\Gamma(i, \lambda)$ coincide. If $\lambda \sim i$, the r paths starting in $F(\lambda)$ end at r distinct points of $F(i)$, hence at all of them; so the path starting at a point of $F(i)$ is one of these paths, read backwards. Hence $i \sim \lambda$ and $p_{i\lambda} = p_{\lambda i}^{-1}$.

(d) Each point of $F(\lambda) = F(i)$ lies on no arc of λ or of i , so it is a path of length 0.

Finally, J_r is a $\mathcal{J}$ -class [4, Proposition 2.1], and $\text{rank}(xy) \leq r$ for $x, y \in J_r$. So by (b) the product of $x, y \in J_r$ in J_r^0 is xy if $\text{bot}(x) \sim \text{top}(y)$ and 0 otherwise, and by (a) and (b) the stated map is an isomorphism. By (c), P has no zero row or column. $\square$

The rows of P are indexed by bottom halves ($\mathcal{L}$ -classes) and the columns by top halves ($\mathcal{R}$ -classes). Proposition 3.1 is the analogue for $\mathcal{B}_n$ and $\mathcal{PB}_n$ of [1, Proposition 8.3] for the partition monoid.

For $\lambda \in H$, let $e_\lambda \in J_r$ be the element with coordinates $(\lambda, 1, \lambda)$, that is, the diagram whose top and bottom halves are both λ and whose through blocks are the pairs $\{a, a'\}$ with $a \in F(\lambda)$. Since $p_{\lambda\lambda} = 1$, Proposition 3.1(b) gives $(\lambda, g, \lambda)(\lambda, h, \lambda) = (\lambda, gh, \lambda)$ for $g, h \in S_r$. So e_λ is an idempotent, and its $\mathcal{H}$ -class $\{(\lambda, g, \lambda) : g \in S_r\}$ is a group isomorphic to S_r .

4 Explicit complete mappings

Throughout this section H is a non-empty finite set, $r \geq 0$, and P is an $H \times H$ matrix over $S_r \cup \{0\}$ with $p_{\lambda\lambda} = 1$ for all λ . We write $\lambda \sim \mu$ when $p_{\lambda\mu} \neq 0$. By Proposition 3.1, the principal factors of $\mathcal{B}_n$ and $\mathcal{PB}_n$ are of this form. We fix a Latin square on H , that is, a map $L: H \times H \rightarrow H$ such that $L(i, \cdot)$ and $L(\cdot, \lambda)$ are bijections for all i, λ ; for instance, identify H with $\mathbb{Z}_N$ and put $L(i, \lambda) = i + \lambda$.

Lemma 4.1. *If S_r has a complete mapping φ , then the map f with $0f = 0$ and $(i, g, \lambda)f = (\lambda, g\varphi, L(i, \lambda))$ is a complete mapping of $\mathcal{M}^0(S_r, H, H, P)$.*

Proof. The map $(i, \lambda) \mapsto (\lambda, L(i, \lambda))$ is a bijection of $H \times H$ and φ is a bijection, so f is a bijection. Since $p_{\lambda\lambda} = 1$, we get $(i, g, \lambda) \cdot (i, g, \lambda)f = (i, g \cdot g\varphi, L(i, \lambda)) \neq 0$, and this is a bijection because $(i, \lambda) \mapsto (i, L(i, \lambda))$ and $g \mapsto g \cdot g\varphi$ are bijections. $\square$

Lemma 4.1 uses only the diagonal of P , as in [1, Proposition 6.5]; the existence statement is a case of (R6).

Now let $r \in \{2, 3\}$. Let $s: S_r \rightarrow \mathbb{Z}_2 = \{0, 1\}$ be the sign homomorphism written additively, with $s(g) = 0$ exactly for even g ; its kernel is the alternating group A_r . We write $\mathcal{M}_2$ for the Rees 0-matrix semigroup $\mathcal{M}^0(\mathbb{Z}_2, H, H, s(P))$, with $\mathbb{Z}_2$ written additively: its non-zero elements are the triples $(i, a, \lambda) \in H \times \mathbb{Z}_2 \times H$, and

$$(i, a, \lambda)(j, b, \mu) = (i, a + s(p_{\lambda j}) + b, \mu) \text{ if } \lambda \sim j, \quad (i, a, \lambda)(j, b, \mu) = 0 \text{ otherwise.}$$

Note that $s(p_{\lambda\lambda}) = 0$.

Lemma 4.2 (lifting). *Let $r \in \{2, 3\}$, and suppose that $\mathcal{M}_2$ has a complete mapping. Then $\mathcal{M}^0(S_r, H, H, P)$ also has a complete mapping.*

Proof. The group A_r has odd order (1 or 3), so $h \mapsto h^2$ is a bijection of A_r and the identity map is a complete mapping of A_r . Since $S_r/A_r \cong \mathbb{Z}_2$ via s , (R5) applies with $N = A_r$. Explicitly, following the proof of [1, Theorem 5.2]: put $g_0 = 1$ and $g_1 = (12)$, and write each $g \in S_r$ as $g = g_u h$ with $u = s(g)$ and $h \in A_r$. If $\bar{f}$ is a complete mapping of $\mathcal{M}_2$ and $(i, u, \lambda)\bar{f} = (j, v, \mu)$, put $(i, g_u h, \lambda)f = (j, p^{-1} h p g_v, \mu)$ with $p = p_{\lambda j}$, and $0f = 0$. Then f is a bijection, and $(i, g_u h, \lambda) \cdot (i, g_u h, \lambda)f = (i, g_u h^2 p g_v, \mu)$; for fixed (i, u, λ) this runs bijectively through the coset of A_r determined by the product $(i, u, \lambda) \cdot (i, u, \lambda)\bar{f}$ in $\mathcal{M}_2$. $\square$

Lemma 4.3. *Let $r \in \{2, 3\}$. Let δ be a permutation of H and $t: H \rightarrow \mathbb{Z}_2$ a map with $t(\mu\delta) = t(\mu) + 1$ for all $\mu \in H$. Define $\bar{f}$ on $\mathcal{M}_2$ by $0\bar{f} = 0$ and, writing $\mu = L(i, \lambda)$,*

$$(i, 1 + t(\mu), \lambda)\bar{f} = (\lambda, 1, \mu), \quad (i, t(\mu), \lambda)\bar{f} = (\lambda, 0, \mu\delta).$$

Then $\bar{f}$ is a complete mapping of $\mathcal{M}_2$. Such δ and t exist when $N = |H|$ is even, so in that case $\mathcal{M}^0(S_r, H, H, P)$ has a complete mapping.

Proof. The element $(\lambda, 1, \nu)$ is the image of exactly one element, coming from the unique i with $L(i, \lambda) = \nu$; the element $(\lambda, 0, \nu)$ is the image of exactly one element, coming from the unique i with $L(i, \lambda)\delta = \nu$. So $\bar{f}$ is a bijection. Since $s(p_{\lambda\lambda}) = 0$, the products are

$$(i, 1 + t(\mu), \lambda)(\lambda, 1, \mu) = (i, t(\mu), \mu), \quad (i, t(\mu), \lambda)(\lambda, 0, \mu\delta) = (i, t(\mu), \mu\delta) = (i, t(\mu\delta) + 1, \mu\delta).$$

For fixed i , $\mu = L(i, \lambda)$ runs through H as λ does, so the products are the elements $(i, t(\nu), \nu)$ and $(i, t(\nu) + 1, \nu)$ for $\nu \in H$, each exactly once. If N is even, let δ be a fixed-point-free involution of H and let t take the values 0 and 1 on the two points of each 2-cycle. The last statement follows from Lemma 4.2. $\square$

Since the diagonal of P is a one-transversal of its pattern, the existence statement of Lemma 4.3 is also a case of (R7); the lemma gives an explicit formula.

Lemma 4.4. *Let $r \in \{2, 3\}$. Let ρ be a permutation of H with $\lambda \sim \lambda\rho$ for all $\lambda \in H$ such that, for every cycle $(\lambda_0, \lambda_1, \dots, \lambda_{m-1})$ of ρ , with indices modulo m ,*

$$m + \sum_{t=0}^{m-1} s(p_{\lambda_t \lambda_{t+1}}) \equiv 0 \pmod{2}. \quad (*)$$

Then there is a map $c: H \rightarrow \mathbb{Z}_2$ with $c(\lambda\rho) = c(\lambda) + 1 + s(p_{\lambda, \lambda\rho})$ for all λ . For such c , the map $\bar{f}$ with $0\bar{f} = 0$ and

$$(i, c(\lambda), \lambda)\bar{f} = (\lambda, 1 + c(\lambda), L(i, \lambda)), \quad (i, 1 + c(\lambda), \lambda)\bar{f} = (\lambda\rho, c(\lambda\rho), L(i, \lambda\rho))$$

is a complete mapping of $\mathcal{M}_2$, and hence $\mathcal{M}^0(S_r, H, H, P)$ has a complete mapping.

Proof. On each cycle of ρ , choose $c(\lambda_0)$ arbitrarily and define $c(\lambda_{t+1}) = c(\lambda_t) + 1 + s(p_{\lambda_t \lambda_{t+1}})$ for $t = 0, \dots, m-2$; condition $(*)$ says exactly that the required relation also holds for $t = m-1$. The images under the first rule are the elements $(\lambda, 1 + c(\lambda), \mu)$, and those under the second rule the elements $(\kappa, c(\kappa), \mu)$ with $\kappa = \lambda\rho$; in both cases μ runs through H as i does. These two sets are complementary, so $\bar{f}$ is a bijection. The products are

$$(i, c(\lambda) + 0 + 1 + c(\lambda), L(i, \lambda)) = (i, 1, L(i, \lambda))$$

from the first rule, which is non-zero since $\lambda \sim \lambda$, and

$$(i, 1 + c(\lambda) + s(p_{\lambda, \lambda\rho}) + c(\lambda\rho), L(i, \lambda\rho)) = (i, 0, L(i, \lambda\rho))$$

from the second, which is non-zero since $\lambda \sim \lambda\rho$. For fixed i they fill $\{i\} \times \{1\} \times H$ and $\{i\} \times \{0\} \times H$ bijectively. The last statement follows from Lemma 4.2. $\square$

Remark 4.5. (1) A fixed point of ρ would be a cycle with $m = 1$ and sum 0, so ρ has no fixed points. (2) If $\lambda \sim \mu$ implies $\mu \sim \lambda$ and $p_{\mu\lambda} = p_{\lambda\mu}^{-1}$, as in Proposition 3.1(c), then every transposition $(a b)$ with $a \sim b$ satisfies $(*)$, because $s(p_{ab}) = s(p_{ba})$. (3) A 3-cycle $(a b c)$ satisfies $(*)$ if and only if $s(p_{ab}) + s(p_{bc}) + s(p_{ca}) = 1$, that is, if and only if $p_{ab}p_{bc}p_{ca}$ is odd. (4) Lemma 4.4 is the special case of the residual-routing criterion [1, Definition 7.13 and Theorem 7.14] in which the normal subgroup is A_r , the routing is $(i, \lambda) \mapsto (\lambda, L(i, \lambda))$ and the residual permutation is $(i, \lambda) \mapsto (i, \lambda\rho)$: the cycles of the residual permutation are the sets $\{i\} \times C$ for the cycles C of ρ , and the cycle condition of [1] becomes $(*)$. We give the direct proof because it yields the explicit formula that is formalised in Section 9.

5 The Brauer monoid

In this section $D = \mathcal{B}_n$, $r \in \{2, 3\}$, $r < n$ and $n \equiv r \pmod{2}$, and $H = H_r(\mathcal{B}_n)$. For an r -subset F of $[n]$, the fibre over F is $H_F = \{h \in H : F(h) = F\}$. It has $m = (n - r - 1)!!$ elements, an odd number, and by Proposition 3.1(d) any two of its elements are compatible.

Lemma 5.1. *$N = |H|$ is odd if and only if $r = 2$ and $n \equiv 2 \pmod{4}$, or $r = 3$ and $n \equiv 3 \pmod{4}$. In these cases $n - r \geq 4$ and $\binom{n}{r}$ is odd.*

Proof. Since m is odd, $N = \binom{n}{r}m$ is odd if and only if $\binom{n}{r}$ is odd. For even n , $\binom{n}{2} = \frac{n}{2}(n - 1)$ is odd if and only if $n \equiv 2 \pmod{4}$. For odd n , the 2-adic valuation of $n(n - 1)(n - 2)$ is that of $n - 1$, so $\binom{n}{3} = n(n - 1)(n - 2)/6$ is odd if and only if $n - 1 \equiv 2 \pmod{4}$, that is, $n \equiv 3 \pmod{4}$. Since $r < n$, this gives $n \geq 6$ for $r = 2$ and $n \geq 7$ for $r = 3$, so $n - r \geq 4$. $\square$

Lemma 5.2 (the triangle). *Suppose that $n - r \geq 2$. Let $E = \emptyset$ if $r = 2$ and $E = \{5\}$ if $r = 3$, and let R_0 be the perfect matching $\{r + 3, r + 4\}, \{r + 5, r + 6\}, \dots$ of $\{r + 3, \dots, n\}$. Put*

$$\begin{aligned} I_0 &= (\{3, 4\} \cup E, \{\{1, 2\}\} \cup R_0), \\ I_1 &= (\{2, 4\} \cup E, \{\{1, 3\}\} \cup R_0), \\ I_2 &= (\{2, 3\} \cup E, \{\{1, 4\}\} \cup R_0). \end{aligned}$$

Then $I_0 \sim I_1$, $I_1 \sim I_2$ and $I_2 \sim I_0$, with $p_{I_0 I_1} = p_{I_1 I_2} = 1$ and $p_{I_2 I_0} = (12)$. In particular $p_{I_0 I_1} p_{I_1 I_2} p_{I_2 I_0}$ is odd.

Proof. The arcs of R_0 are common to the three half-diagrams and give double edges, and the point 5 (if $r = 3$) is free in all three and is the largest free point, so it is joined to itself. Only the points $1, \dots, 4$ matter. In $\Gamma(I_0, I_1)$ the path from 3 is $3, 1, 2$ (arcs $\{1, 3\}$ of I_1 and $\{1, 2\}$ of I_0) and ends at $2 \in F(I_1)$, and 4 is fixed; the order is preserved. In $\Gamma(I_1, I_2)$, 2 is fixed and the path from 4 is $4, 1, 3$; the order is preserved. In $\Gamma(I_2, I_0)$ the path from 2 is $2, 1, 4$ and 3 is fixed, so the smaller free point 2 of I_2 is joined to the larger free point 4 of I_0 , and $p_{I_2 I_0} = (12)$. $\square$

These are the three half-diagrams used for $\mathcal{P}_n$ in the proof of [1, Theorem 8.5], where the marked singletons become free points, completed to perfect matchings by R_0 . With the idempotents e_λ defined after Proposition 3.1, Lemma 5.2 says that $e_{I_0} e_{I_1} e_{I_2} e_{I_0} = (I_0, p_{I_0 I_1} p_{I_1 I_2} p_{I_2 I_0}, I_0)$ is an odd element of the group $\mathcal{H}$ -class of e_{I_0} .

Lemma 5.3 (joining two fibres). *Let $F \neq F'$ be r -subsets of $[n]$ with $F \setminus F' = \{x_1, \dots, x_k\}$ and $F' \setminus F = \{y_1, \dots, y_k\}$, and suppose that $r + 2k \leq n$. Then there are $v \in H_F$ and $w \in H_{F'}$ with $v \sim w$.*

Proof. Choose distinct points $z_1, \dots, z_k \notin F \cup F'$, which is possible since $|F \cup F'| + k = r + 2k \leq n$, and a perfect matching R of the remaining $n - r - 2k$ points, an even number. Put

$$v = (F, \{\{z_t, y_t\} : t \leq k\} \cup R), \quad w = (F', \{\{x_t, z_t\} : t \leq k\} \cup R).$$

In $\Gamma(v, w)$ the path from x_t is x_t, z_t, y_t and ends in F' , the points of $F \cap F'$ are paths of length 0, and the arcs of R give double edges. So $v \sim w$. $\square$

Lemma 5.4 (matching). *Suppose that $n - r \geq 4$ and $\binom{n}{r}$ is odd, and let $X = \{I_0, I_1, I_2\}$ be as in Lemma 5.2. Then $H \setminus X$ can be partitioned into pairs $\{v, w\}$ with $v \sim w$.*

Proof. The elements of X lie in the three distinct fibres over $\{3, 4\} \cup E$, $\{2, 4\} \cup E$ and $\{2, 3\} \cup E$, one in each. The other $\binom{n}{r} - 3$ fibres, an even number, can be paired so that each pair (F, F') satisfies $r + 2|F \setminus F'| \leq n$. Indeed, if $r = 2$ then $|F \setminus F'| \leq 2$ and $n \geq 6$, and if $r = 3$ and $n \geq 9$ then $|F \setminus F'| \leq 3$; in these cases any pairing works. If $(n, r) = (7, 3)$ the condition is

$F \cap F' \neq \emptyset$, and the following 16 pairs of 3-subsets of $[7]$ work: $\{a, b, 6\}$ with $\{a, b, 7\}$ for the ten 2-subsets $\{a, b\}$ of $[5]$; $\{1, 6, 7\}$ with $\{1, 2, 3\}$, $\{2, 6, 7\}$ with $\{1, 2, 4\}$, $\{3, 6, 7\}$ with $\{1, 3, 4\}$, $\{4, 6, 7\}$ with $\{2, 3, 4\}$ and $\{5, 6, 7\}$ with $\{1, 2, 5\}$; and $\{1, 3, 5\}$ with $\{1, 4, 5\}$. These are exactly the 32 subsets other than $\{3, 4, 5\}$, $\{2, 4, 5\}$ and $\{2, 3, 5\}$, and the two sets in each pair meet. For each pair of fibres, Lemma 5.3 gives compatible $v \in H_F$ and $w \in H_{F'}$, and we take $\{v, w\}$ as a pair. Now exactly one element of each fibre has been used, either in X or in a pair. The remaining $m - 1$ elements of each fibre, an even number, are pairwise compatible, and we pair them arbitrarily. $\square$

Proof of Theorem 1.1(a). Let $r < n$ be a rank that occurs in $\mathcal{B}_n$. By Proposition 3.1, $J_r^0 \cong \mathcal{M}^0(S_r, H, H, P)$ with $p_{\lambda\lambda} = 1$ for all λ . If $r \notin \{2, 3\}$, then S_r has a complete mapping by (R1), and Lemma 4.1 applies. If $r \in \{2, 3\}$ and N is even, Lemma 4.3 applies. If $r \in \{2, 3\}$ and N is odd, then $n - r \geq 4$ and $\binom{n}{r}$ is odd by Lemma 5.1. Let ρ be the permutation of H that maps $I_0 \mapsto I_1 \mapsto I_2 \mapsto I_0$ and interchanges the two elements of each pair given by Lemma 5.4. Then $\lambda \sim \lambda\rho$ for all λ . The 3-cycle satisfies $(*)$ by Lemma 5.2 and Remark 4.5(3), and the transpositions satisfy it by Proposition 3.1(c) and Remark 4.5(2). So Lemma 4.4 gives a complete mapping of J_r^0 . $\square$

6 The partial Brauer monoid

Lemma 6.1. *The number $a(k)$ of involutions of a k -set is even for $k \geq 2$. Hence, if $n \geq 4$, $r \in \{2, 3\}$ and $r < n$, then $N = \binom{n}{r}a(n - r)$ is even.*

Proof. We have $a(2) = 2$, $a(3) = 4$ and $a(k) = a(k - 1) + (k - 1)a(k - 2)$, so $a(k)$ is even for $k \geq 2$ by induction. If $n - r \geq 2$ this gives the claim; otherwise $n - r = 1$, so $(n, r) = (4, 3)$ and $N = 4$. $\square$

Proposition 6.2. *The principal factor of rank 2 of $\mathcal{PB}_3$ is isomorphic to the Brandt semigroup $B(S_2, 3) = \mathcal{M}^0(S_2, [3], [3], \Delta_3)$ and has no complete mapping.*

Proof. The half-diagrams of rank 2 are the three pairs $(F, \emptyset)$ with $|F| = 2$, the third point being a singleton. If $F(\lambda) \neq F(i)$, a point of $F(\lambda) \setminus F(i)$ is a singleton of i , hence an isolated vertex of $\Gamma(\lambda, i)$ outside $F(i)$, so $\lambda \not\sim i$. With Proposition 3.1(c), the sandwich matrix is the identity matrix Δ_3 over S_2 . It is obtained from the normalised 3×3 matrix all of whose entries are the identity, which has odd order, by replacing the off-diagonal entries by 0. Since $S_2 \cong C_2$ has a non-trivial cyclic Sylow 2-subgroup and 3 is odd, (R8) shows that there is no complete mapping. $\square$

The same conclusion follows from [1, Theorem 10.1], since $B(S_2, 3)$ is an inverse semigroup whose non-zero $\mathcal{J}$ -class has three $\mathcal{L}$ -classes and maximal subgroup S_2 . An exhaustive computer search confirms it (Section 8).

Proof of Theorem 1.1(b). Let $r < n$. If $r \notin \{2, 3\}$, Lemma 4.1 applies as in the proof of part (a). If $r \in \{2, 3\}$ and $n \geq 4$, then N is even by Lemma 6.1, and Lemma 4.3 applies. If $r \in \{2, 3\}$ and $n \leq 3$, then $(n, r) = (3, 2)$, and Proposition 6.2 applies. $\square$

7 The whole monoids

Proof of Theorem 1.1(c). The group of units of $\mathcal{B}_n$ and of $\mathcal{PB}_n$ is $J_n \cong S_n$, so (i) and (ii) each imply (iii) by (R3). Statements (iii) and (iv) are equivalent by (R1). Suppose that (iii) holds, and let φ be a complete mapping of S_n . The principal factor of the group of units is S_n with a zero adjoined, and φ extended by $0 \mapsto 0$ is a complete mapping of it. Every proper principal

factor of $\mathcal{B}_n$ has a complete mapping by part (a). Every proper principal factor of $\mathcal{PB}_n$ has one by part (b), since $n \neq 3$. So (i) and (ii) hold by (R4). $\square$

8 Computations

No statement of this paper depends on a computation: each has a written proof above. The computations are independent checks of these proofs and of our conventions. The programs (in Python), certificates and logs are in the directory `certificates/brauer-complete` of version 1.4.0 of the repository [3].

Two implementations. The first implementation multiplies diagrams with a union-find structure, computes the Rees coordinates of Proposition 3.1, and builds complete mappings exactly as in the proofs (Lemmas 4.1–4.4 and 5.2–5.4, with $L(i, \lambda) = i + \lambda$ after indexing H by $\mathbb{Z}_N$). The second implementation shares no code with the first: it has its own parser, multiplies diagrams by following alternating paths through the middle row, enumerates the elements itself, and never uses Rees coordinates. A certificate is a text file with one line $x \mapsto x\alpha$ for every element x of a principal factor (without the zero, which is fixed) or of a whole monoid. For a factor J_r^0 , the second implementation checks that the listed x are valid diagrams of rank r , pairwise distinct, and $N^2 r! = |J_r|$ in number, so that they are exactly the elements of J_r ; that the values $x\alpha$ are pairwise distinct and of rank r ; and that the products $x \cdot x\alpha$ have rank r (that is, are non-zero in J_r^0) and are pairwise distinct. For a whole monoid it checks that the listed x are all the elements, that α is a bijection, and that $x \mapsto x \cdot x\alpha$ is a bijection. Table 2 lists the certificates; all passed.

Certificate for	elements	construction
$J_2^0(\mathcal{B}_6)$, $N = 45$	4050	Lemmas 4.4, 5.2, 5.4 and 4.2; the first case with N odd
$J_3^0(\mathcal{B}_7)$, $N = 105$	66150	the same; the first case with N odd and $r = 3$
$\mathcal{B}_6$	10395	ranks 0, 2, 4, 6; rank 2 as above
$\mathcal{B}_1, \mathcal{B}_4, \mathcal{B}_5$	1, 105, 945	Lemmas 4.1, 4.2 and 4.3
$\mathcal{PB}_6$	140152	ranks 0 to 6; ranks 2 and 3 by Lemma 4.3
$\mathcal{PB}_1, \mathcal{PB}_4, \mathcal{PB}_5$	2, 764, 9496	Lemmas 4.1, 4.2 and 4.3
$J_2^0(\mathcal{B}_6)$, $J_3^0(\mathcal{B}_7)$	4050, 66150	second versions: a triangle with sign bits (0, 1, 0) found by search and a maximum matching computed with the NetworkX library in place of Lemmas 5.2 and 5.4

Table 2: Certificates. The number of elements excludes the zero of a principal factor. Each certificate was built by the first implementation and checked by the second.

For the whole monoids, the factors of rank at least 4 and the groups of units need complete mappings of S_4 , S_5 and S_6 . Those of S_4 and S_5 were found with the SAT solver CaDiCaL, and one of S_6 by an exact-cover search (Knuth’s Algorithm X) in about seven minutes, after CaDiCaL had not finished in twenty minutes. They are checked implicitly by the whole-monoid certificates. For the theorems themselves, (R1) supplies these complete mappings.

Further checks.

- *Rees coordinates.* The first implementation confirms that the $\mathcal{J}$ -classes are the rank classes for $\mathcal{B}_n$ with $n \leq 5$ and $\mathcal{PB}_n$ with $n \leq 4$, and that Proposition 3.1(a)–(c) and the product rule (b) hold for all pairs in J_r for these monoids. For $\mathcal{B}_6$ the product rule was checked for all pairs in ranks 0 and 6 and for 400 000 random pairs in each of ranks 2 and 4; for $\mathcal{PB}_5$, for all pairs in the ranks with $|J_r|^2 \leq 4 \cdot 10^6$ and for 400 000 random pairs in each other rank. The values of N agree with the formulas of Section 3.

- *The odd case.* For $(n, r) = (6, 2), (7, 3), (10, 2), (11, 3)$ and $(14, 2)$, that is, for $N = 45, 105, 4725, 17325$ and 945945 , the first implementation confirms the sandwich permutations of Lemma 5.2 and that the construction of Lemma 5.4 yields a partition of $H \setminus X$ into compatible pairs. The second implementation confirms, without Rees coordinates, that for $n \in \{6, 7, 10, 11\}$ the product $e_{I_0} e_{I_1} e_{I_2} e_{I_0}$, with e_λ as defined after Proposition 3.1, is the element of the group $\mathcal{H}$ -class of e_{I_0} that interchanges the through strands at the points 3 and 4.
- *Parities.* The parity pattern of Table 1 holds for all $n \leq 200$ in both families.
- *Non-existence.* An exhaustive search, using the second implementation, confirms that S_2 and S_3 have no complete mapping, that the units of $\mathcal{B}_2, \mathcal{B}_3, \mathcal{PB}_2$ and $\mathcal{PB}_3$ are the permutation diagrams, and that the 19-element rank-2 principal factor of $\mathcal{PB}_3$ has no complete mapping (a depth-first search over all bijections, with 67 911 890 nodes). As positive controls, the same search finds complete mappings of three other small principal factors.

One script reruns all these checks and regenerates the certificates; on its last run (26 September 2026, 07:42–07:44 UTC) every certificate was reproduced with the same SHA-256 hash, and the hashes are recorded in the repository.

9 Formal verification

The Lean file `lean/Research/BrauerCMCore.lean` of version 1.4.0 of the repository [3] (324 lines) formalises Lemmas 4.3 and 4.4 at the level of the semigroup $\mathcal{M}_2$, for an arbitrary finite index set. It uses Lean 4 (version 4.33.1) and Mathlib [6] (commit `0df444a3`). The declarations are in the namespace `BrauerCMCore`. The assertion of Lemma 4.3 that $\bar{f}$ is a complete mapping of $\mathcal{M}_2$ is the theorem `BrauerCMCore.lemmaB`, and the corresponding assertion of Lemma 4.4 is the theorem `BrauerCMCore.lemmaC`. The following listing is an excerpt of this file, not the complete file. It shows the definitions of `Elt`, `mul` and `IsCompleteMapping` and the statements of `lemmaC` and `lemmaB`, each inside its section together with the `variable` declaration of that section. Proofs, auxiliary lemmas, comments and the definitions of `fC` and `fB` are omitted, and some long lines are broken differently.

```
abbrev Elt (H : Type*) := Option (H × ZMod 2 × H)
variable {H : Type*}
def mul (P : H → H → Option (ZMod 2)) : Elt H → Elt H → Elt H
| none, _ => none
| some _, none => none
| some (i, a, l), some (j, b, m) => (P l j).map (fun p => (i, a + p + b, m))
def IsCompleteMapping {S : Type*} (op : S → S → S) (α : S → S) : Prop :=
  Function.Bijective α ∧ Function.Bijective (fun x => op x (α x))

section LemmaC
variable (P : H → H → Option (ZMod 2)) (L : H → H → H) (ρ : H ≃ H)
(c s : H → ZMod 2)
theorem lemmaC [Finite H] (hdiag : ∀ l, P l l = some 0)
(hρ : ∀ l, P l (ρ l) = some (s l)) (hc : ∀ l, c (ρ l) = c l + 1 + s l)
(hrow : ∀ i, Function.Injective (L i))
(hcol : ∀ l, Function.Injective (fun i => L i l)) :
  IsCompleteMapping (mul P) (fC L ρ c)
end LemmaC
```

```

section LemmaB
variable (P : H → H → Option (ZMod 2)) (L : H → H → H) (δ : H ≃ H)
  (t : H → ZMod 2)
theorem lemmaB [Finite H] (hdiag : ∀ l, P l l = some 0)
  (hδ : ∀ μ, t (δ μ) = t μ + 1)
  (hrow : ∀ i, Function.Injective (L i))
  (hcol : ∀ l, Function.Injective (fun i => L i l)) :
  IsCompleteMapping (mul P) (fB L δ t)
end LemmaB

```

Here `none` is the zero of $\mathcal{M}_2$, `some (i, a, l)` is the element (i, a, λ) , and `P l j = none` means $\lambda \not\sim j$. The maps `fC` and `fB` send `none` to `none` and are the maps of Lemmas 4.4 and 4.3: `fC` sends `some (i, a, l)` to `some (l, 1 + c l, L i l)` if $a = c l$, and to `some (ρ l, c (ρ l), L i (ρ l))` otherwise; `fB` sends it to `some (l, 1, L i l)` if $a = 1 + t (L i l)$, and to `some (l, 0, δ (L i l))` otherwise. The predicate `IsCompleteMapping` is the definition of a complete mapping of a magma; associativity is not used.

The Lean statements differ from Lemmas 4.3 and 4.4 as follows. The matrix is over $\mathbb{Z}_2$ directly, with no reference to S_r or to diagrams; only its diagonal entries (`hdiag`) and, for `lemmaC`, its entries at $(\lambda, \lambda\rho)$ (`hρ`) are constrained. In `lemmaC` the map c is a hypothesis; that such a c exists under condition $(*)$ is the first step of the proof of Lemma 4.4 and is not formalised. The Latin square is given by the hypotheses `hrow` and `hcol`.

The following are not formalised: Proposition 3.1 (diagram monoids are not in Mathlib); the lifting Lemma 4.2, that is, (R5); Lemma 4.1; Lemmas 5.1–5.4, 6.1 and Proposition 6.2; the results (R1)–(R8) of [1] and the results of [4] that we use; and hence Theorem 1.1 itself. The Lean proof certifies the two constructions of complete mappings over $\mathbb{Z}_2$ for every finite index set, in particular for every N ; it does not certify that the hypotheses hold for the principal factors of $\mathcal{B}_n$ and $\mathcal{PB}_n$.

The file contains no `sorry` and no axiom declarations, and does not use `native_decide`; the ten uses of `decide` are closed statements about $\mathbb{Z}_2$ with at most three variables. The module was built with `lake build Research.BrauerCMCore` on 26 September 2026 (exit code 0). The axiom audits (`#print axioms`) of `lemmaC` and `lemmaB` report only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The compiled module was then checked with `leanchecker` (exit code 0), which replays its declarations in Lean’s kernel and trusts the imported Mathlib; this is not an independent kernel. The code and logs are available at [3].

Tool and computational resource disclosure

This work was carried out on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents; the agents selected this problem. Claude Code agents found the proofs, wrote the programs and the Lean formalisation, carried out the literature searches described in the introduction, and drafted this text. The programs, logs and Lean code are available at [3]. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. The correctness of the results rests on the written proofs, which use the results of [1] and [4] listed in Section 2; the Lean formalisation covers only the two constructions described in Section 9, and the computations of Section 8 are additional checks. AI systems are not authors of this paper; the author takes full responsibility for its content.

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